Response to the qualifying committee

When Pork-Barrel Backfires: Coalition Presidentialism, Polarization, and the Price of Legislative Support in Brazil

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June 2026

Overview

This document summarizes how each of the feedback items raised by the qualifying committee (Daniel Cajueiro, Bernardo Mueller, and Rafael Terra, defense of 02 June 2026) was addressed in the post-defense revision. We organize the responses around the four blocks of the internal `MUSTDO_v2.md` roadmap: empirical extensions (Block A), writing improvements (Block B), external-data integration (Block C), and direct responses to committee questions (Block D). Each entry indicates the source of the comment (committee member or our own notes), the original concern, and the implementation in the revised manuscript. The final draft is `docs/paper.pdf` in this repository.

1 Block A — Empirical extensions

A.1 Polarization terciles separately by legislature

[Bernardo Mueller; Question 3 from the committee]
Comment: "Terciles in the pooled analysis do not allow us to distinguish 'polarization causes backfire' from 'the Bolsonaro era is the phenomenon'."
Response: The tercile analysis has been re-cut within each legislature in the revised draft (Subsection 6.3, Table 10). The within-legislature pattern is informative: the Weak Divergence measure produces a U-shape that survives the legislature split, and the Strong and Euclidean measures replicate the cross-legislature gap inside Legislature 56 alone. The pooled tercile result reported in the qualifying-defense draft has been superseded by the within-legislature estimates and is no longer the basis for the central interpretation.

A.2 Structural break in February 2021 (Lira presidency)

[Our notes, in light of committee discussion]
Comment: "Under Bolsonaro there is a massive expansion of the secret budget and Arthur Lira is elected Chamber president. This changes everything."
Response: The structural break in February 2021 is documented explicitly in the revised draft as the *Legislative Capture* subsection (Subsection 6.4, Table 11, Figure 5). The Lira sub-period analysis is the empirical core of the post-defense reframing: pork did not stop functioning under Bolsonaro; the principal of the bargain shifted from the Executive to the Centrão-led Chamber leadership. Three sub-samples (Maia 55, Maia 56, Lira 56) and two exclusion sub-samples (excl. PP, excl. Centrão) localize the institutional transfer.

A.3 Centrão variable and heterogeneity

[Bernardo Mueller]
Comment: "Maybe the effect you attribute to polarization is in fact the Centrão swallowing the amendments."
Response: The Centrão indicator and the corresponding heterogeneity analysis are reported in Subsection 6.5 (Table 12). The sub-sample analysis confirms that the post-Lira Centrão alignment effect is concentrated within Centrão deputies themselves; once Centrão deputies are removed from the post-Lira sub-sample, the estimate is reduced to non-significance. The Centrão composition list follows the standard nine-party categorization of Power (2010) and is documented in the table notes.

A.4 Alternative outcome: alignment with the Centrão

[Our notes]
Response: Implemented in Subsection 6.5 (Table 12). The exercise was generalized in two directions during the post-defense revision: alignment with the broader Centrão (the original A.4) and alignment with the party of the Chamber president (the new Subsection 6.4, Table 11). The two specifications jointly support the Legislative Capture interpretation.

A.5 Quadratic treatment specification

[Committee question 3]
Response: The quadratic specification is implemented as a robustness exercise in Appendix F (Table 14). The original Double Machine Learning estimator with two simultaneously endogenous variables encountered numerical instability arising from the near-collinearity of T and T^2 in the ElasticNet nuisance regression; we therefore implemented the augmented model via classical IV-2SLS with a reduced control set (eleven covariates surviving QR-based rank validation), within-deputy demeaning, and deputy-clustered standard errors. The estimates indicate a mild U-shaped response in all three samples with the inflection located near $ R$3 million per deputy-vote observation. Because the median T is approximately R$0.8 million and the 90th percentile is below R$3 million, the curvature is observed only in the upper tail and does not affect the qualitative interpretation of the linear estimates in Table 4.

A.6 Two-round PEC analysis

[Bernardo Mueller]
Response: The within-PEC analysis is reported in Appendix G (Table 15). We identified nine constitutional amendments with both first-round and second-round votes recorded in our panel and estimated the within-PEC change in alignment as a function of the within-PEC change in committed amendment value, with PEC fixed effects. The sample is small (3,383 deputy-PEC observations), and the precision of the estimates is limited; we report the exercise as a within-vote corroboration of the broader pattern rather than as an independent identification strategy.

A.7 Stylized facts and aggregate descriptives

[Bernardo Mueller; Our notes]
Comment: "You have to show the picture: base versus Centrão, who receives more in proportional terms."

Response: The new Subsection 2.4 ("Aggregate stylized facts") consolidates the descriptive material: annual amendment commitments by primary-result code 2014–2025 (Table 1, with the post-ADPF 854 substitution from RP-9 to RP-8 and Pix amendments visible in the row-by-row comparison), the composition stacked-area chart (Figure 2), and Centrão versus non-Centrão alignment and allocation patterns by sub-period (Table 2). The subsection draws on the federal-budget data described in our response to Block C.

A.8 Consolidated diagnostic test table

[Bernardo Mueller]
Response: Implemented as Table 3 (`tab:iv_validation`). The table reports first-stage F, Anderson–Rubin 95% CI, Sargan–Hansen J (p-value), and Cinelli–Hazlett $\mathrm{RV}_{q=1}$ in a single panel separated by legislature, allowing integrated assessment of identification strength.

A.9 Weak vs. Strong Divergence as mediator

[Our notes; companion polarization paper]
Response: The dual-mediation analysis is reported in Appendix H (Table 16). The two-mediator Acharya–Blackwell–Sen system shows that the Strong Divergence channel transmits 31.4% of the Temer-era effect and the Weak Divergence channel transmits a negligible share. The exercise corroborates the single-mediator analysis in the main text and isolates Strong Divergence as the relevant transmission mechanism, consistent with the credibility-cost interpretation developed in Section 7.

A.9.5 Legislative offices as a parallel pork channel

[Post-defense discussion]
Response: The legislative-office heterogeneity analysis is reported in Subsection 6.6 (Table 13). Deputies holding Tier-2 positions (party leadership, committee presidency or vice-presidency, formal rapporteur appointment) display a Bolsonaro-era backfire sharply attenuated relative to rank-and-file deputies. The pattern is the observational counterpart of the Legislative Capture interpretation: deputies with access to procedural levers are insulated from the credibility cost that drives the rank-and-file backfire.

2 Block B — Writing improvements

B.1 Following the literature's conventions

[Committee question 1]
Comment: "On page 10 you have to follow the template of the literature more closely."
Response: The empirical-strategy and identification sections have been rewritten with the conventions of recent Public Choice and EJPE papers (notation, ordering of identification arguments, placement of diagnostic tests). The specific page-10 concern is flagged for a final pass with Bernardo before submission.

B.2 Move descriptive text into tables and figures

[Committee comment]

Response: The revised draft replaces a substantial portion of the previously prose-based descriptive content with tables (1–16, with seven new tables added post-defense) and figures (1–6). The seasonality figure, the legislative-capture forest plot (Figure 5), and the quadratic-response curve (Figure 7) are illustrative examples.

B.3 Justification of the control set

[Bernardo Mueller]
Response: Section 4 contains a new paragraph documenting the 142-variable full-clean control set, grouped into five families, with the justification for the wide set resting on (i) the sparsity-after-selection property of DML, (ii) the multidimensional Brazilian institutional environment, and (iii) the ablation experiment reported in Appendix B (Table B.13) showing that the structural coefficient is essentially flat across alternative specifications.

B.4 Reformulating the backfire interpretation

[Our notes]
Response: Section 7 (Discussion) was substantially rewritten. The reformulation has three layers: (i) the IV strategy rules out the reverse-causality interpretation by construction; (ii) the credibility-cost mechanism (a deputy receiving a visible amendment incurs a public-legibility cost under polarized constituencies); and (iii) the Legislative Capture finding shows that the substantive content of pork-for-votes survives but is re-routed through a different institutional actor.

3 Block C — External data integration

C.1 Federal-budget aggregates and category composition

[Committee question 2]
Response: Table 1 of the revised paper reports the 2014–2025 federal-budget aggregates by primary-result category, sourced from the Federal Transparency Portal and cross-validated against the Federal Treasury open-data CKAN repository.

C.2 Mandatory execution legal framework

[Committee question 3]
Response: The new Subsection 2.1 ("The legal architecture of mandatory execution") discusses Constitutional Amendments 86 of 2015, 100 of 2019, and 105 of 2019 explicitly.

C.3 Deputy-level RP-9 (Secret Budget) data

[Committee question 4]
Response: The deputy-level RP-9 indicator d_{rp9} was constructed from the public list of *apoiamentos* released under ADPF 854 (Ofícios 932/2024 and 114/2025) and is incorporated as a control in the augmented specifications reported in Table 9 (the dual-outcome IV-DML estimates with and without opaque-channel proxies). An imputed RP-9 measure based on the prefeito-level GitHub data (Proxy P4 in `STATE_OF_PLAY.md` §1.5.bis) is used in the legislative-capture discussion in Subsection 6.4. The official disclosure is incomplete, and we acknowledge this limitation explicitly in Section 7.

C.4 Post-STF substitution: RP-8 and Pix amendments

[Our notes]
Response: The post-2022 substitution dynamics from RP-9 to RP-8 and Pix amendments are documented in Subsections 2.2 and 2.4 (Table 1, Figure 2). The Transparência Brasil et al. (2025) technical note on continued ADPF 854 violations is cited in Sections 2.2 and 7.

4 Block D — Direct responses to committee questions

D.1 Reverse causality

Question: "Can you test for reverse causality?"
Response: The IV strategy rules out reverse causality by construction (the orthogonalization step removes variation in T_{it} correlated with deputy-specific unobservables). We additionally report placebo tests (random Bernoulli outcome, post-vote amendment treatment) in Section 5, all of which produce coefficients indistinguishable from zero.

D.2 ADPF 850 (sic ADPF 854) as a natural experiment

Question: "Do you exploit the natural experiment provided by the STF intervention?"
Response: The event study around the November 2021 STF preliminary injunction on RP-9 is reported in Appendix A (`app:event`, Figure 6). We deliberately use the preliminary injunction rather than the final December 2022 ruling to avoid confounding the budget intervention with the simultaneous change of federal administration on 1 January 2023. The event study is descriptive (no exogenous control group exists) and reported as supplementary evidence rather than as causal identification.

D.3 Exclusion of Legislature 57

Question: "Why do you exclude Legislature 57?"
Response: Two reasons, documented in Subsection 2.2. First, the official amendment-execution data from the Federal Transparency Portal exhibits a structural lag of one to two years; the last fully-closed budget exercise at the time of writing is 2024. Second, Legislature 57 falls inside the post-ADPF 854 substitution regime (RP-8 and Pix amendments), which is institutionally distinct from the RP-9 era and would mix heterogeneous regimes in the panel.

D.4 Typology of pork-barrel regimes

Question: "Do your findings support a typology of regimes?"
Response: Yes. The Introduction (Section 1) and the Discussion (Section 7) propose a two-regime typology: **Regime A** (programmatic, cooperative-coalition; Temer 2016–2018) and **Regime B** (populist-polarized, legislatively-captured; Bolsonaro 2019–2022). The two regimes differ in the sign of the visible-amendment coefficient, in the identity of the principal of the bargain, and in the salience of opaque parallel channels.

Summary

Of the 22 substantive items in the post-defense roadmap, 22 have been implemented in the revised manuscript. The remaining item (B.1, a final stylistic pass against the literature's page-10 conventions) is flagged for a last review with Bernardo before submission to Public Choice. The post-defense reframing reorganizes the central narrative around the Legislative Capture finding (Subsection 6.4): the pork-for-votes coefficient does not disappear under populist polarization; it changes hands. We are grateful for the committee's careful reading and for the substantive direction the post-defense feedback provided.